CHRISTOPHE ABBOUD

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PROFESSIONAL SUMMARY

Mechanical Engineer and M2 student in Mechatronics, Artificial Intelligence and Robotic Systems at Sorbonne Université, with research experience in wearable rehabilitation technologies, sensing, mechatronic systems, and experimental robotics. Experienced in experimental development, embedded data acquisition, signal processing, and mechanical design, with research publications in intelligent manufacturing and sensing. Interested in human–robot interaction, assistive robotics, robot control, and predictive human-motion modeling.

EDUCATION

SORBONNE UNIVERSITÉ — **PARIS, FRANCE**

Master 2: Mechatronics, Artificial Intelligence and Robotic Systems — *Sept. 2026 - PRESENT*

- Specializing in robotic systems, with a focus on collaborative robotics and human–robot interaction
- Relevant coursework: wearable robotics, haptics and teleoperation, physical human robot interactive control

LEBANESE AMERICAN UNIVERSITY — **BYBLOS, LEBANON**

Bachelor of Engineering in Mechanical Engineering — *Sept. 2022 - JUNE. 2026*

- CGPA: 3.97/4.00: Named on Dean's distinguished list - Recipient of USAID HES Scholarship - Honors Program student
- Graduated 1st in the Mechanical Engineering class and received the Charbel Khairallah Endowed Engineering Excellence Award

COLLEGE SAINT JOSEPH — **AKKAR, LEBANON**

Lebanese Baccalaureate: General Science — *JUNE. 2022*

- Grade: 19.167/20 - Ranked 2nd nationally - Mention Très Bien

RESEARCH EXPERIENCE

LEBANESE AMERICAN UNIVERSITY — **BYBLOS, LEBANON**

Research Assistant - Wearable Rehabilitation & Sensing Technologies — *SEP.2024 – AUG.2026*

- Designed and prototyped a compact, 3D-printable wearable rehabilitation device for 3-finger motion tracking, integrating soft pneumatic sensing with VR-based feedback using an ESP32 C-Mini
- Developed embedded data-acquisition and signal-processing for real-time analysis of pneumatic sensor signals
- Conducted iterative experimental testing and data analysis to evaluate sensing performance and motion-tracking capability

Research Assistant – Piezoelectric micro surface texturing — *SEP.2024 – AUG.2025*

- Designed and developed a novel piezoelectric tool for CNC micro-machining, reaching repeatable surface textures of $50\ \mu m$
- Contributed to an AI vision-based neural-network approach for automated surface-quality assessment of the textures

PUBLICATIONS

LEBANESE AMERICAN UNIVERSITY — **BYBLOS, LEBANON**

- Bejjani R, AlLabaky C, Abboud C. Piezo device for micro-texturing surfaces with vision-based neural network quality assessment. Science Progress. 2026;109(1) doi:10.1177/00368504261436958
- Abboud, C. et al. *"Tool Design for Micro-Machining Using a Piezo-Actuated System"* – Accepted for presentation at the 18th International Conference on High-Speed Machining, Metz, France, 2025
- Al Fleety H., Abboud C., Al Labaki C., Bejjani R., "Effect of Micro-Dimple Texturing and Solid Lubricants (Graphene and MoS2) on the Tribological Performance of Aluminum Surfaces under Line-Contact Conditions", International Journal of Advanced Manufacturing Technology, 2025, Under review
- Tawk C., Abboud C., Naous S., Nasr B., Wearable 3D-Printed Soft Pneumatic Sensors for AR driven Rehabilitation. In Progress

RELEVANT PROJECTS

LEBANESE AMERICAN UNIVERSITY — **BYBLOS, LEBANON**

Assistive Robot for Visually Impaired Navigation — *Apr.* 2024

- Designed and manufactured robotic navigation prototype for visually impaired users, integrating ultrasonic and infrared sensors, torque sensors, and a NI myRIO controller
- Developed and implemented the robot's sensing and control architecture for detection and navigation assistance

AI-Enhanced Electrical Power Steering System — *Sep.2025 – May.2026*

- Designed and programmed an AI-enhanced Electrical Power Steering system, integrating mechanical, electrical, and control components with fuzzy logic and PID control
- Implemented and validated the steering control logic using MATLAB/Simulink and Texas Instruments LaunchXL controller

UNIVERSITY OF CORSE **CORSE, FRANCE**

- Ranked 1st in the 5th EUROMED worldwide scientific high school competition (CPIE-RINASCITA-certified). May. 2021

PROFESSIONAL EXPERIENCE

DZINE PLUS – PRODUCT DESIGN INTERN **BEIRUT, LEBANON**

- Assisted in STEM kit product design and prototyping, integrating controllers and embedded systems *DEC. 2025 – FEB.2026*

DAR AL HANDASAH – SIDARA COMPANY **BEIRUT, LEBANON**

- Gained hands-on experience in control processes within the Intelligent Systems and Automation Unit *Jul. 2025 - AUG. 2025*
- Assisted in producing 15+ Piping & Instrumentation Diagrams (P&IDs) and control diagrams using AutoCAD.

MIDDLE EAST AIRLINES – MEAG GSEM **BEIRUT, LEBANON**

- Gained hands-on experience with automotive subsystems, engines, diagnostics, and safety procedures. *Jun. 2025*
- Assisted in maintenance, inspection, troubleshooting, and documentation of airport ground support equipment.

MERCEDES BENZ - T. GARGOUR & FILS S.A.L. **BEIRUT, LEBANON**

- Assisted in vehicle diagnosis, troubleshooting, maintenance and engine disassembly/reassembly. *May. 2025*
- Worked with electrical connections and vehicle control units.

LEBANESE AMERICAN UNIVERSITY **BYBLOS, LEBANON**

- Tutored students in Control Systems, Dynamics, and Material Properties & Processes *SEP. 2023 – JUN. 2026*

BEYMAN – REMOTE **SYDNEY, AUSTRALIA**

- Training in REVIT software. Modeled and detailed various HVAC systems and Piping networks *JUN.2023 - DEC.2023*

TECHNICAL SKILLS

- **Programming and simulation**: Python, GitHub, MATLAB/Simulink, Java, LabVIEW, EES, Unity (AR/VR), Microsoft Office
- **Robotic and Control**: Robotic Manipulator Control, PID Control, Control Systems, Real-Time Control
- **Embedded Systems**: Microcontrollers, NI myRio, TI LaunchXL, Sensor Integration, Data Acquisition, Signal Processing
- **AI & Data:** Neural Networks, Fuzzy Logic, Machine Learning
- **Mechanical Design & Manufacturing:** SolidWorks (CSWA), AutoCAD, Revit, 3D Printing (FDM), and CNC Machining, Ansys

VOLUNTEERING EXPERIENCE AND SOCIAL ACTIVITIES

YOUTH GROUP - MOUVEMENT APOSTOLIQUE MARIALE **AKKAR, LEBANON**

- Active and dedicated member and volunteer with more than 40 hours per month *2010 – PRESENT*
- Leader of a youth group of 12-16 years old, leading and organizing weekly meetings *2023 – 2026*
- Planning and executing youth camps and village events, including fundraising initiatives, and workshops, demonstrating leadership, initiative, and strong organizational skills. *2020 – 2026*

LEBANESE AMERICAN UNIVERSITY **BYBLOS, LEBANON**

- ASME Member: Assisted in organizing on-campus and off-campus events, including industrial site visits *SEP. 2022 - JUN. 2026*
- Engineering summer camp: Assisted high school students in the engineering summer camp for 5 weeks *JUN. 2024 - JUL. 2024*
- USAID-Mentor: Guided two mechanical engineering students in academic and personal development *SEP. 2024 - JUN. 2025*
- USAID-Community service project: Afforestation awareness and distributing agricultural tools in Chouf *SEP. 2024 - DEC. 2024*
- President of the Millenium Fellowship International project at LAU leading more than 20 students *SEP.* 2023 - JUN. 2024

LANGUAGES

Arabic: Native **English:** C1 Advanced **French**: B2 – DELF Certified

HOBBIES

Piano – Hiking